

Dictionary

It is collection of keys and values.

Properties

```
In [ ]: 1)Mutable
        2)Ordered/Unordered
        3)We can not use indexing and slicing.
        4)Key can not be duplicate
        5)Values can be duplicates.
        6)Enclosed by {}
        7)Initialize dictionary by using dict() or {}

Python2->Unordered
Python 3->Ordered

Data Types of key:-Immutable(int,float,complex,str,tuple,)

Data Types of value:-Any Data type(int,float,complex,str,list,tuple,set,dict)

dict1=
```

Syntax:

```
In [ ]: dict1={key:Value}

dict2={}=dict()  #dictionary initialization
```

```
In [2]: d1={"a":97,"b":98,"c":122}
        d1
```

```
Out[2]: {'a': 97, 'b': 98, 'c': 122}
```

```
In [3]: d1={"a":97,"b":98,"c":122}
        d1["a"]
```

```
Out[3]: 97
```

```
In [4]: d1={"a":97,"b":98,"c":122}
        d1["B"]
```

```
-----
KeyError                                Traceback (most recent call last)
Cell In[4], line 2
      1 d1={"a":97,"b":98,"c":122}
----> 2 d1["B"]

KeyError: 'B'
```

```
In [5]: d2={"a":33,"b":22,"c":[111,222,333]}
d2
```

```
Out[5]: {'a': 33, 'b': 22, 'c': [111, 222, 333]}
```

```
In [6]: d2={"a":33,"b":22,"c">{111,222,333}}
d2
```

```
Out[6]: {'a': 33, 'b': 22, 'c': {111, 222, 333}}
```

```
In [7]: len(d2)
```

```
Out[7]: 3
```

Dictionary Functions

1.get()

```
In [ ]: dict.get("key")
```

```
In [9]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1['A']
```

```
Out[9]: 100
```

```
In [11]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1.get('A')
```

```
Out[11]: 100
```

2)keys()

```
In [ ]: dict.keys()
```

```
In [12]: dict1={'A':100,'B':200,'C':300,'D':400}
dict1.keys()
```

```
Out[12]: dict_keys(['A', 'B', 'C', 'D'])
```

```
In [13]: dict1={'A':100,'B':200,'C':300,'D':400}
for i in dict1.keys():
    print(i)
```

```
A
B
C
D
```

```
In [15]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
len(employee_data)
```

```
Out[15]: 4
```

```
In [16]: employee_data.get('Designation')
```

```
Out[16]: 'Data Scientist'
```

```
In [17]: employee_data.keys()
```

```
Out[17]: dict_keys(['Name', 'Age', 'Designation', 'Salary'])
```

```
In [20]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
for key in employee_data.keys():
    print("key is:",key)
    print("Values is",employee_data[key])
    print('*'*40)
```

```
key is: Name
Values is Yash
*****
key is: Age
Values is 22
*****
key is: Designation
Values is Data Scientist
*****
key is: Salary
Values is 70000
*****
```

3)values()

```
In [ ]: Syntax:Dict.values()
```

```
In [21]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
employee_data.values()
```

```
Out[21]: dict_values(['Yash', 22, 'Data Scientist', 70000])
```

```
In [22]: employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}
for value in employee_data.values():
    print("Values is:",value)
    print('*'*40)
```

```
Values is: Yash
*****
Values is: 22
*****
Values is: Data Scientist
*****
Values is: 70000
*****
```

4)items()

In []: Syntax: Dict.items()

In [23]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`employee_data.items()`

Out[23]: dict_items([('Name', 'Yash'), ('Age', 22), ('Designation', 'Data Scientist'), ('Salary', 70000)])

In [24]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`for i in employee_data.items():`
 `print(i)`

('Name', 'Yash')
 ('Age', 22)
 ('Designation', 'Data Scientist')
 ('Salary', 70000)

In [26]: `a,b=('Name', 'Yash')`
`a`

Out[26]: 'Name'

In [27]: `b`

Out[27]: 'Yash'

In [28]: `employee_data={'Name':'Yash','Age':22,'Designation':"Data Scientist",'Salary':70000}`
`for key,value in employee_data.items():`
 `print(f"key=={key} and value=={value}")`
 `print("*"*40)`

key==Name and value==Yash

 key==Age and value==22

 key==Designation and value==Data Scientist

 key==Salary and value==70000

In []: